

ABHAY SAHARAWAT

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PROFESSIONAL SUMMARY

AI & Data Science undergraduate with hands-on experience building full-stack, AI-integrated applications and computer vision systems in **Python**, **JavaScript**, **React**, and **Node.js**. Skilled in **Machine Learning**, **Data Analysis**, and **REST API** development, with a track record of shipping working software end-to-end — from model integration and backend services to responsive front-end interfaces. Seeking an AI Engineer, Python/Full-Stack Developer, or Data Analyst role to apply strong analytical thinking and engineering fundamentals to production-quality, data-driven products.

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript, SQL, HTML, CSS

AI/ML & Data Analysis: Machine Learning, Computer Vision, Generative AI Integration (Gemini API), NumPy, Pandas, Matplotlib, Scikit-learn, OpenCV, YOLO

Full-Stack / Web: React, Vite, Node.js, Express.js, REST APIs, Tkinter (GUI), HTML, CSS

Databases & Tools: SQL, Git/GitHub, Microsoft Excel, PowerPoint, Word

Core CS Concepts: Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, Cloud Computing, Cryptography, Heuristic & Entropy Analysis

PROJECTS

ResumeIQ — AI-Powered Resume Analyzer

July 2026

React, Vite, Node.js, Express.js, Gemini AI

- Built a full-stack resume evaluation platform using **React**, **Vite**, **Node.js**, and **Express.js**, integrating the **Gemini AI API** to deliver **ATS scoring** and personalized improvement feedback.
- Engineered secure **PDF upload and parsing** pipeline to extract resume content for automated AI analysis.
- Generated structured **AI feedback** across Skills, Projects, Experience, and Formatting to guide resume optimization before job applications.
- Designed a responsive, modern **dashboard UI/UX** and managed API credentials securely via environment variables.

Real-Time Traffic Density Estimation System

May 2026

Python, OpenCV, YOLOv8

- Built a real-time vehicle detection and lane-wise traffic monitoring system using **YOLOv8** and **OpenCV**.
- Implemented **multi-lane vehicle counting** and traffic intensity classification (Smooth/Heavy) with live bounding-box and statistics overlays.
- Processed live video streams for continuous, **low-latency traffic surveillance** and analysis.

PassGenX — Entropy-Based Password Security Tool

Oct 2025

Python, Cryptography

- Developed a **cryptographically secure** password generator using strong random-function primitives to eliminate predictable patterns.
- Designed an **entropy-based password strength analyzer** to evaluate resistance to brute-force attacks.
- Implemented optional **local encryption** for secure storage of user credentials.

Lightweight Malware Scanner

Nov 2024

Python, Tkinter, hashlib

- Built a malware detection system applying **SHA-1 hashing** and **heuristic analysis** to flag suspicious files.
- Developed a **Tkinter-based GUI** for real-time file scanning, quarantine of infected files, and log generation.

EDUCATION

Graphic Era Deemed to be University, Dehradun, India

2023 – 2026

BCA, Specialization in Artificial Intelligence & Data Science — CGPA: 6.8/10

CERTIFICATIONS & ACHIEVEMENTS

- Advanced Machine Learning Course** — Coursera (2025)
- Human-Computer Interaction (Elite)** — NPTEL, IIT Madras
- Deloitte Technology Job Simulation** — Forge
- TATA GenAI-Powered Data Analytics Job Simulation** — Forge